

DIY Activity: Customer Order Behavior Analysis Using E-commerce Data

Guidelines for Students: This is a self-paced Do-It-Yourself (DIY) activity, and **submissions are not required**. The primary objective is to enhance your skills through practical, hands-on learning. Engage in discussions with peers to deepen your understanding and collaboratively work through the activity. If you have any questions or need clarification, **please ask them during the upcoming live sessions** for a more comprehensive explanation.

Overview: A retail company wants to better understand its customers' purchasing patterns across different states, product categories, and payment modes. By analyzing historical orders and detailed transaction information, the company aims to refine its marketing strategies, optimize inventory, and tailor payment offers to specific customer segments.

Datasets:

Orders Dataset

- **Order ID:** A unique identifier for each order.
- **Order Date:** The date when the order was placed.
- **Customer Name:** The name of the customer placing the order.
- **State:** The state in which the customer resides.
- **City:** The city where the customer is located.

Details Dataset

- **Order ID:** A unique identifier matching the order with the "Orders.csv" dataset.
- **Amount:** The total amount of the order.
- **Profit:** The profit generated from the order.
- **Quantity:** The quantity of products ordered.
- **Category:** The broad category to which the product belongs.
- **Sub-Category:** A more specific sub-category of the product.
- **Payment Mode:** The mode of payment used for the order.

Tasks to be performed:

1. Analyze the relationship between the amount and the profit on the orders:

- Calculate the correlation coefficient between these two variables to determine the strength and direction of the relationship using the CORREL function.
- Create a scatter plot to visualize the relationship between Amount and Profit on the orders.

Identify if higher-order amounts result in higher profits and discuss possible business implications.

2. Explore how Category impacts the Profitability:

- Create a box plot to show the distribution of Profit across each Category, highlighting the spread, median, and any outliers.
- Use descriptive statistics such as mean and median to compare profitability between categories.

Identify the most profitable category and suggest reasons for its performance based on margin, sales volume, or cost structure.

(Hint: Merge the datasets using Order ID and use a box plot to visualize profit per category.)

3. Analyze Payment Preferences by State:

- Use a pivot table to count the number of orders per Payment Mode across different States.
- Visualize the data using a stacked bar chart to compare preferences regionally.

Interpret which payment modes are favored in which states and suggest how this could guide location-specific offers.

(Hint: Use VLOOKUP to merge the datasets using Order ID before analysis.)

4. Investigate the relationship between Quantity and Profitability:

- Create a scatter plot with Quantity on the X-axis and Profit on the Y-axis to visualize the trend.
- Analyze whether higher quantities lead to proportionally higher profit or if there are diminishing returns.

Use the results to suggest optimal order sizes and pricing strategies for maximum profitability.

(Hint: Filter out extreme outliers if needed to focus on common trends.)

5. Study City-Wise Spending Behavior:

- Create a pivot table to calculate the total Amount spent by customers in each City.
- Visualize the top 10 cities by total spending using a bar chart.

Identify high-revenue cities and propose localized campaigns to boost engagement and sales further.

(Hint: Use conditional formatting to highlight the top cities in your table view.)

6. Analyze Monthly Trends in Order Volumes:

- Extract the Month and Year from the Order Date to group orders chronologically.
- Count the number of orders placed each month to understand trends over time.

Discuss seasonal or promotional effects observed in order behavior.

7. Determine the Most Popular Product Sub-Categories:

- Use a pivot table to count the total number of orders for each Sub-Category.
- Create a pie chart to visualize the proportion of each sub-category within total orders.

Identify the top-selling sub-categories and recommend bundling or promotional strategies to boost related sales.

(Hint: Consider filtering the chart to show only the top 5–10 sub-categories for clarity.)

8. Examine State-wise Profit Margins:

- Calculate the average Profit per order for each State using a pivot table.
- Use a heatmap or column chart to visualize profit margins geographically.

Highlight states with low average profits and suggest region-specific solutions like pricing adjustments or product focus.

9. Correlation Between Order Quantity and Total Order Amount:

- Use the CORREL function to measure the strength of the relationship between Quantity and Amount.
- Create a scatter plot to visually assess whether larger orders in quantity lead to higher order value.

Discuss how strong the correlation is and how it impacts inventory planning and fulfillment strategies.

(Hint: Remove zero or outlier values to improve the accuracy of your correlation analysis.)

10. Impact of Payment Mode on Average Order Value:

- Use a pivot table to calculate the average Amount for each Payment Mode.
- Create a bar chart to visually compare the average order values across modes.

Use the insights to recommend discounts or rewards targeting high-value payment options.

(Hint: Sort the chart from highest to lowest to easily spot the most valuable payment methods.)

Suggestion:

- Merge Orders.csv and Details.csv using Order ID as a key (you can use VLOOKUP or INDEX MATCH).
- Use Excel pivot tables, filters, and charts (scatter, bar, line, box) for analysis and visualization.